CAPSTONE · TRACK A · INFRASTRUCTURE-FIRST

Staging-first delivery for kk-payments

Every release is built, deployed to staging, smoke-tested and approved before it reaches production. A failed release rolls itself back.

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The gap

Until now, every kk-payments release was run **by hand from a laptop**, straight into the only environment we had: production.

One environment

No safe place to test a release against production-like settings.

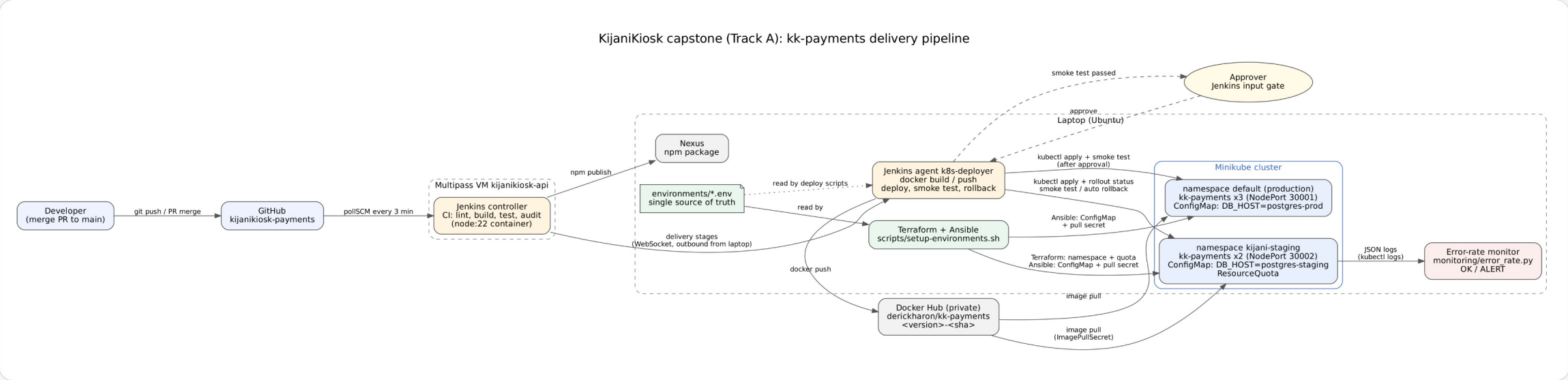
Config in the manifest

Settings hardcoded; logs unstructured; "Ready" only meant "started".

Rollback needs a human

Week 9: a bad image stalls the rollout until someone notices at 02:00.

Architecture



Terraform creates the kijani-staging namespace and caps its resources.

Ansible puts each environment's settings and registry login into its namespace.

Jenkins moves each release through staging to an approval gate, then production.

The pipeline, build #48

CI
Lint, test, audit, npm publish

Image
Build, push 1.1.0-94a2b5a

Staging
Deploy, wait for rollout

Smoke test
Version, environment, 202, 400, request ID in logs

Approval
A human gate, shown only after smoke passes

Production
Same image, same scripts, smoke again

> **What the approver sees:** SMOKE PASSED: environment=staging version=1.1.0-94a2b5a requestId=smoke-staging-1794405619

Key decision: never drop capacity

Not chosen: maxUnavailable 1

My Week 9 watch log showed the rollout terminating an old Pod immediately: **2 of 3 Pods** served for about 12 seconds.

Chosen: maxUnavailable 0, maxSurge 1

An old Pod is removed only after its replacement passes the readiness probe on /health. Production stayed at 3 of 3 throughout.

Trade-off: one extra Pod during every rollout, and a broken release stalls instead of degrading. That stall is exactly what the pipeline turns into an automatic rollback.

A real failure: build #47

1. Deployed	2. Smoke failed	3. Rolled back	4. Prod untouched
1.1.0-f4af50b rolled out to staging	Payment request timed out (curl exit 28)	Staging back on 1.1.0-036ed12, automatically	Approval and production stages skipped

Root cause: the smoke test ran while old Pods were still terminating, so a request hit a dying Pod. **Fix (build #48):** wait for old Pods to be gone and retry each smoke request. Then the release passed.

AI tooling, reviewed

Used for	Got right	Got wrong	I changed
Jenkinsfile	Rollback runs once, behind a flag	Smoke piped into tee under /bin/sh: a failed test would pass	bash with pipefail
Smoke test	Checks version, environment, request ID	grep -q under pipefail gave a false failure; no retry during rollout	Read logs first; wait for old Pods; retries
Scope draft	Specific, measurable criteria	Contradicted Track A: kijani-prod, no monitoring	Rewrote against the guide

Every entry is in **docs/ai-governance-log.md** in the eight-field format. Each mistake was caught by running the code, not by trusting it.

Before real customer traffic

Component	Gap	Fix
k8s-deployer agent	Deploys with my cluster-admin kubeconfig	ServiceAccount per namespace with a scoped RBAC Role
Minikube	One laptop node runs both environments	Managed cluster, 3+ nodes; staging and production apart
Container user	Named user, so runAsNonRoot cannot be enforced	Numeric UID 10001 plus runAsNonRoot
Error-rate monitor	Runs per build or on demand, not continuously	Prometheus alert routed to on-call

Descoped: the Week 10 receipt chain. Its integration point would be a receipts-bucket setting in the same ConfigMap.